WEB TRAFFIC ANALYTICS DASHBOARD

Traffic channels • Engagement • Conversion funnel • Drop-off analysis

2,400	40.8%	92s	6.1%
Sessions	Bounce rate	Avg. session duration	Conversion rate

Project objective

Analyze website traffic sources, engagement, and conversion behavior to identify where visitors leave the journey and recommend practical improvements to website performance.

Dataset note: No source dataset was attached with the task. This report uses a clearly labelled synthetic dataset of 2,400 sessions for demonstration and submission formatting. Replace the sample data with the downloaded dataset before presenting the findings as real business results.

Prepared as a student project report • October 2026

1. Executive Summary

The sample contains **2,400 sessions**, with an overall bounce rate of **40.8%**, average session duration of **92 seconds**, and an overall conversion rate of **6.1%**. Average pageviews per session are **2.56**. These measures give a baseline for comparing channels and identifying user-experience problems.

Key observations from the illustrative data

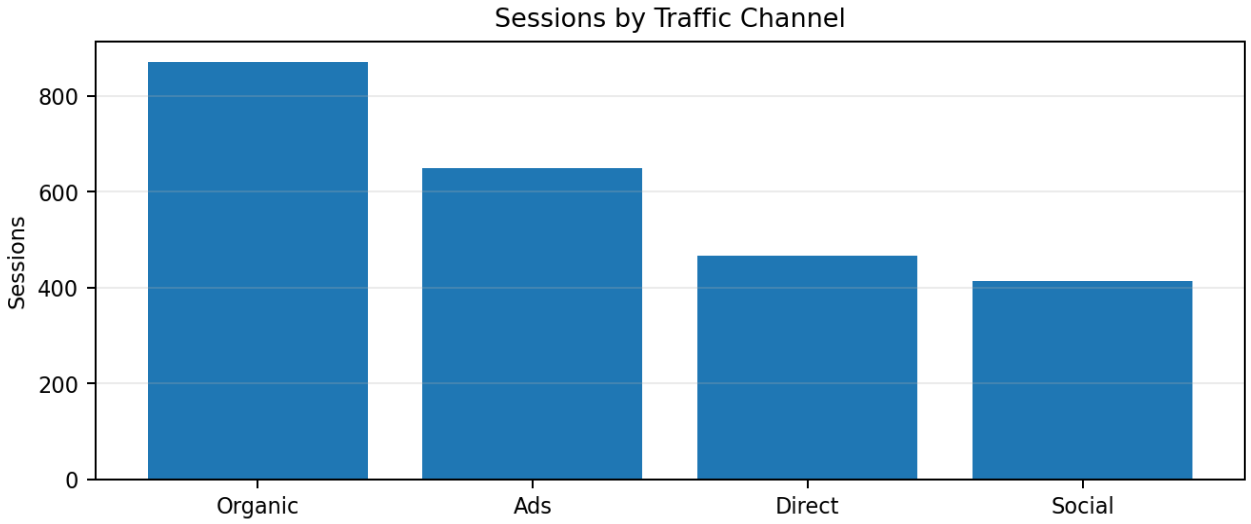
• **Direct** has the highest conversion rate (9.0%). Review its landing pages and user journey for patterns worth reusing.

• **Social** has the highest bounce rate (57.6%), suggesting a need to check message match, page speed, and mobile experience.

• **Direct** has the longest average session duration (111 seconds), which may indicate stronger engagement.

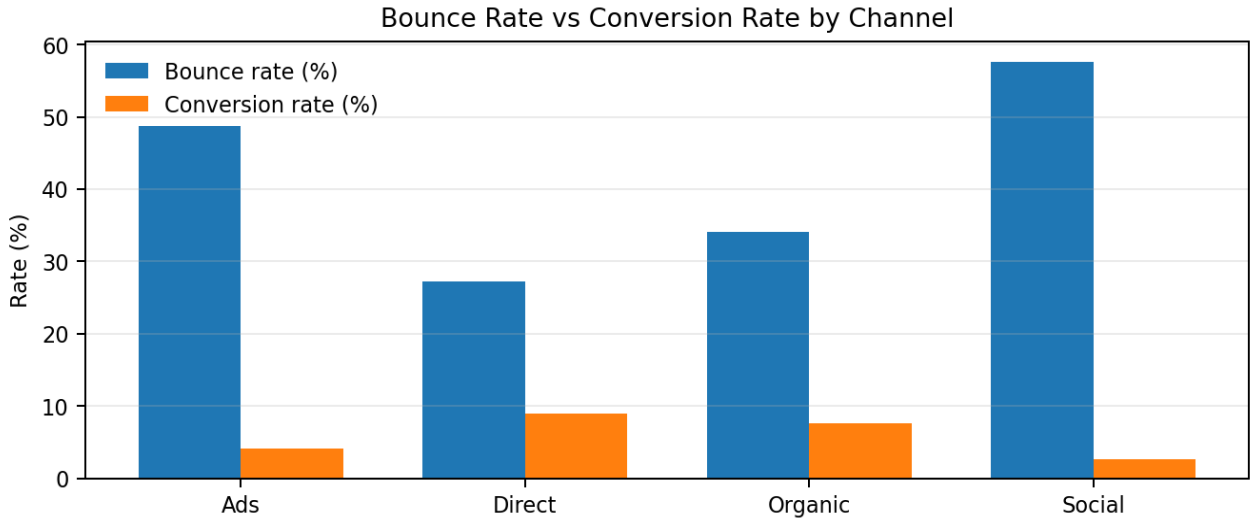
Interpret these as demonstration findings only; generated data is not evidence about an actual website.

2. Traffic Channel Performance



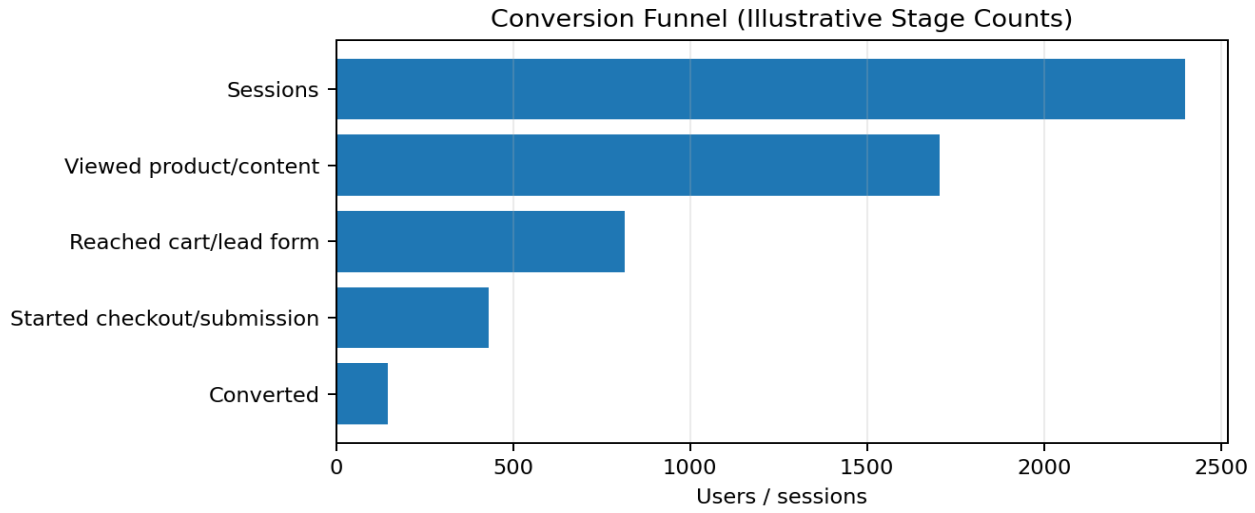
Channel	Sessions	Bounce rate	Avg duration	Pages/session	Conversions	Conv. rate
Ads	649	48.7%	79s	2.39	27	4.2%
Direct	467	27.2%	111s	2.93	42	9.0%
Organic	871	34.1%	101s	2.74	66	7.6%
Social	413	57.6%	69s	2.05	11	2.7%

3. Engagement and Conversion



Bounce rate measures the share of sessions that leave after viewing only one page (definition can vary by analytics platform). Average session duration is the mean time spent per session. Pageviews per session are used here as a practical proxy for page consumption; the task says “per user cohort,” so in a real dataset this should be calculated by user or cohort ID if available.

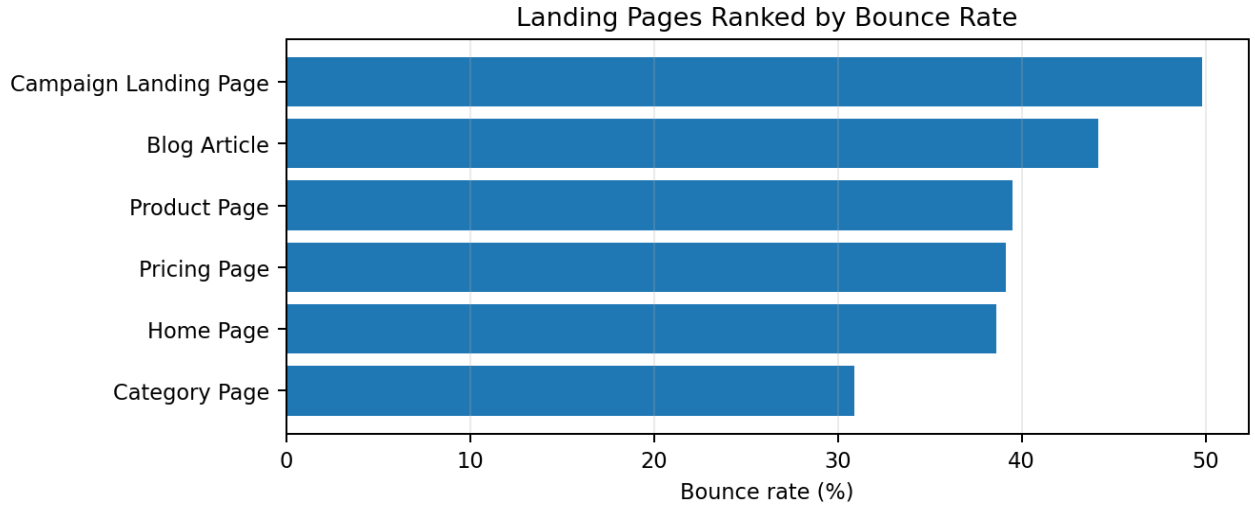
4. Conversion Funnel and Drop-off



Funnel stage	Users/sessions	Drop-off from previous
Sessions	2,400	—
Viewed product/content	1,704	29.0%
Reached cart/lead form	816	52.1%
Started checkout/submission	432	47.1%
Converted	146	66.2%

The funnel stages are illustrative estimates, not event counts observed from a real tracking system. In a production analysis, count unique users reaching each event in order (landing → product/content → cart/lead form → checkout/submission → purchase/lead) and calculate drop-off between consecutive stages.

5. Landing Page Drop-off Analysis



Landing page	Sessions	Bounce rate	Avg. session duration
Campaign Landing Page	297	49.8%	76s
Blog Article	421	44.2%	85s
Product Page	696	39.5%	90s
Pricing Page	189	39.2%	95s
Home Page	632	38.6%	98s
Category Page	165	30.9%	116s

A high bounce rate is a signal to investigate, not proof that a page is bad. Blog posts may satisfy visitors in one view, while product or campaign pages are expected to guide visitors to another action. Review bounce rate alongside conversion, engagement time, device, and landing-page intent.

6. Recommendations

1. Improve high-bounce landing pages: Check whether the page headline and call-to-action match the traffic source's promise. Make the next step visible without excessive scrolling.

2. Optimize paid campaigns: Compare ad keywords/creative with landing-page content. Pause or revise campaigns that attract clicks but produce few qualified conversions.

3. Strengthen mobile usability: Since mobile often accounts for a large share of visits, test page speed, responsive layouts, form fields, and checkout on real mobile devices.

4. Reduce funnel friction: Shorten forms, show delivery/pricing information early, make errors understandable, and offer guest checkout or a simple lead form where appropriate.

5. Measure outcomes: Use conversion rate, qualified leads/revenue, and completion by device/channel—not traffic volume alone—to prioritize changes.

6. Run controlled experiments: A/B test one major change at a time and evaluate results over a suitable sample size.

7. Methodology and Formulas

The synthetic session-level dataset was generated to demonstrate the requested calculations. Each row represents one session and contains traffic channel, device, landing page, duration, pageviews, bounce flag, conversion flag, and last-stage label.

Bounce rate = (Bounced sessions ÷ Total sessions) × 100

Average session duration = Total session duration ÷ Number of sessions

Pageviews per session = Total pageviews ÷ Total sessions. For pageviews per user cohort, group sessions by user ID or cohort and calculate the metric at that level.

Conversion rate = (Converted sessions ÷ Total sessions) × 100. If the business measures users instead of sessions, use unique converted users ÷ unique users.

Funnel drop-off = ((Users at previous stage – Users at next stage) ÷ Users at previous stage) × 100

Suggested data columns for the real dataset

Column	Purpose
session_id / user_id	Identify sessions and deduplicate users
date / timestamp	Trend analysis and cohorting
source / medium	Classify Organic, Ads, Social, Direct
landing_page / page_path	Find page-level engagement and exits
session_duration	Calculate average time spent
pageviews	Measure content consumption
conversion_event	Mark purchase, signup, or qualified lead
funnel_event / event_timestamp	Reconstruct ordered conversion steps
device_category	Compare mobile, desktop, and tablet

8. Limitations

- Results are based on synthetic demonstration data because no dataset was provided.
- The funnel counts are illustrative and should be replaced with event-level counts from the actual analytics export.
- Session duration and bounce definitions vary across tools such as GA4 and other analytics platforms.
- Attribution can be affected by cross-device journeys, consent settings, ad blockers, and channel-tagging quality.
- The report does not claim statistical significance or actual revenue impact.

9. Conclusion

A useful web traffic dashboard combines acquisition (sessions by channel), engagement (bounce rate, duration, pageviews), and outcomes (conversion rate and funnel completion). The sample report shows how to compare channels, locate pages that deserve review, and turn findings into testable improvements. For final submission as a real analytics result, replace the synthetic sample with the downloaded dataset and refresh every metric and chart.

Appendix: Sample dataset fields used in this report are synthetic and intended for demonstration only.